Golemio
Data Platform

Operátor ICT, a.s.
2025-12-15

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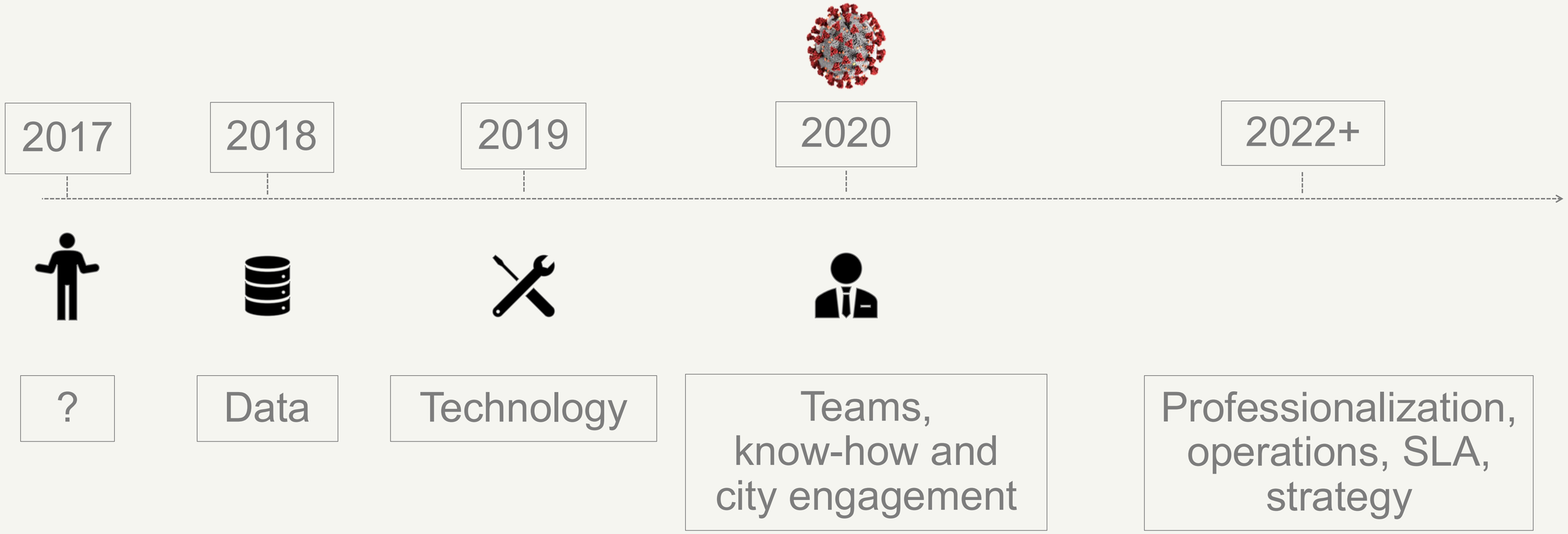
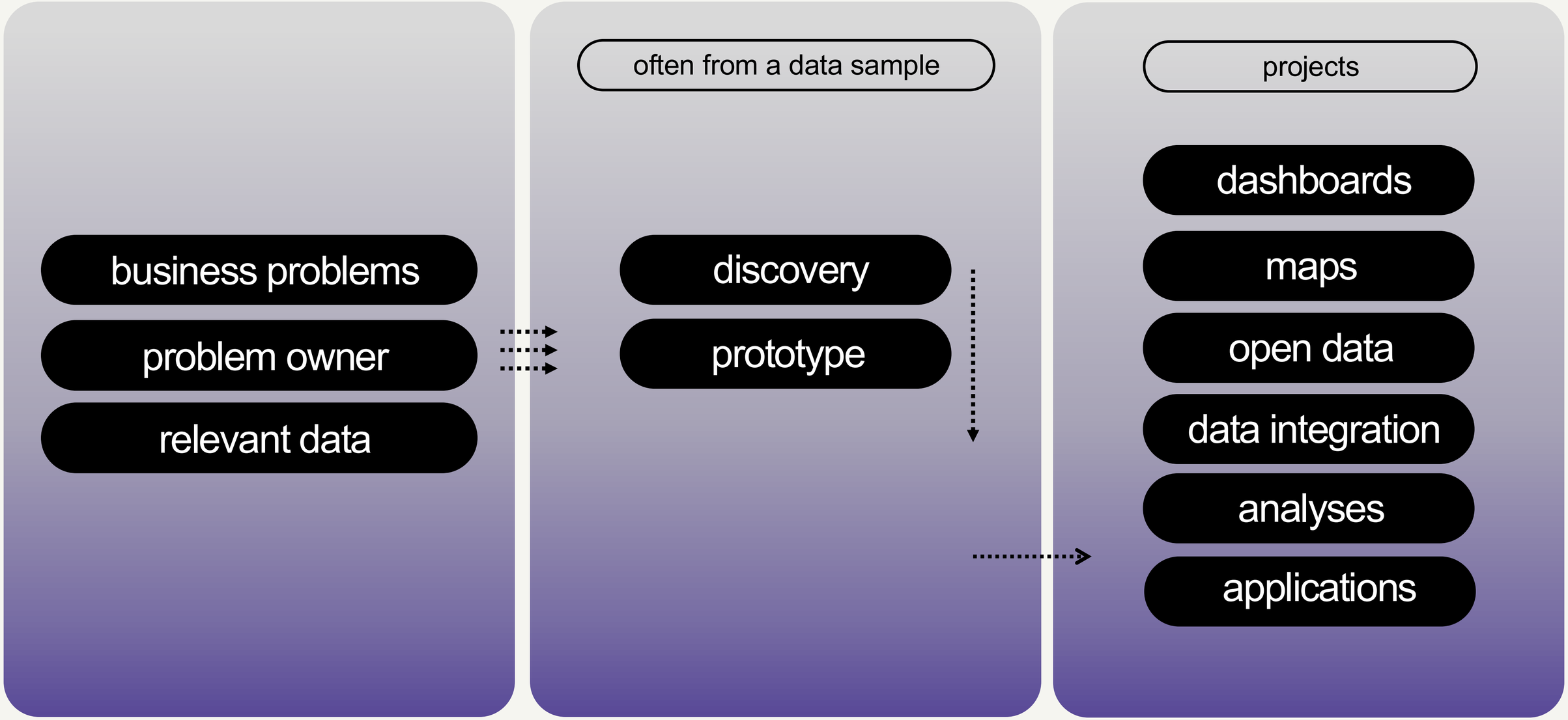
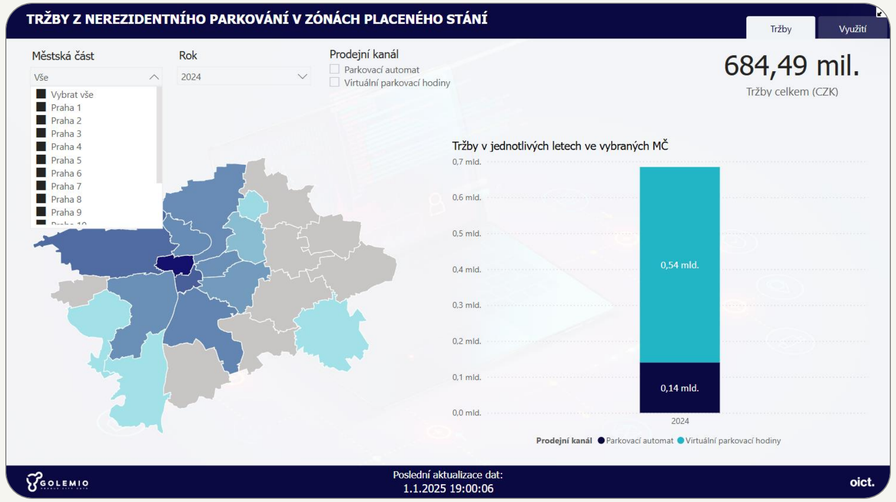
01 Golemio


2017
2018
2019
2020
2022+
?
Data
Technology
Teams, know-how and city engagement
Professionalization, operations, SLA, strategy

01 Golemio


business problems
problem owner
relevant data
often from a data sample
discovery
prototype
projects
dashboards
maps
open data
data integration
analyses
applications

01 Golemio



- We deliver data to users through **user-friendly, interactive dashboards and maps**
- Registered users can access dashboards via the **Golemio BI** web interface
- Selected dashboards are **publicly available** after free registration at https://bi.golemio.cz.
- Dashboards created in MS Power BI, maps use vector tiles (MVT).

Examples of publicly available dashboards

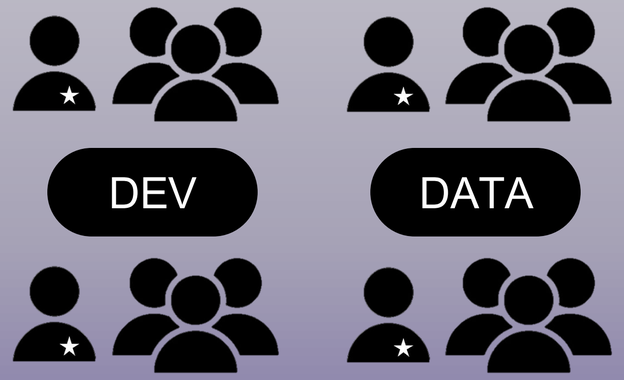
- **Traffic travel time indicators** (Waze data), GBI.
- **P+R parking occupancy** in Prague (TSK data via API), GBI.
- **Paid parking zones – non-resident payments overview**, GBI.

02 We Are Not a Tool, We Are a Team

consultants

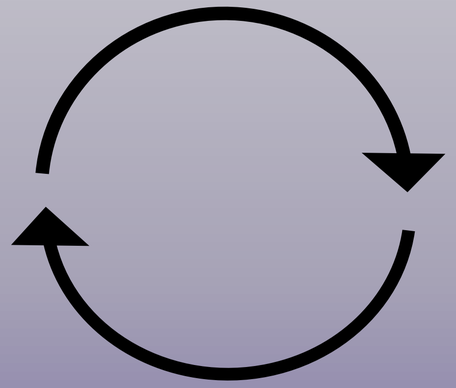
account manager, project manager

4 delivery teams



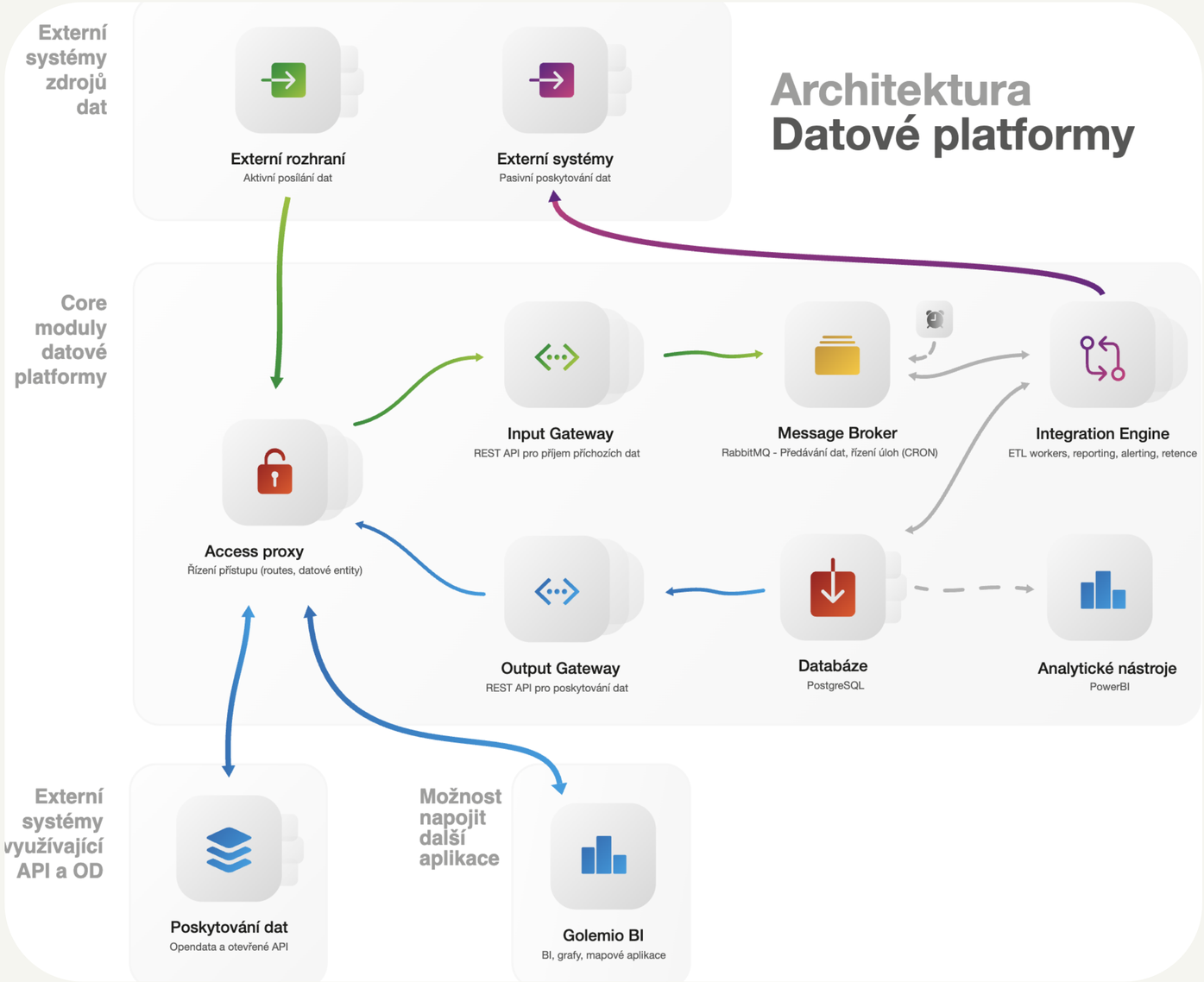
lead by product owner

sprint-based deliveries



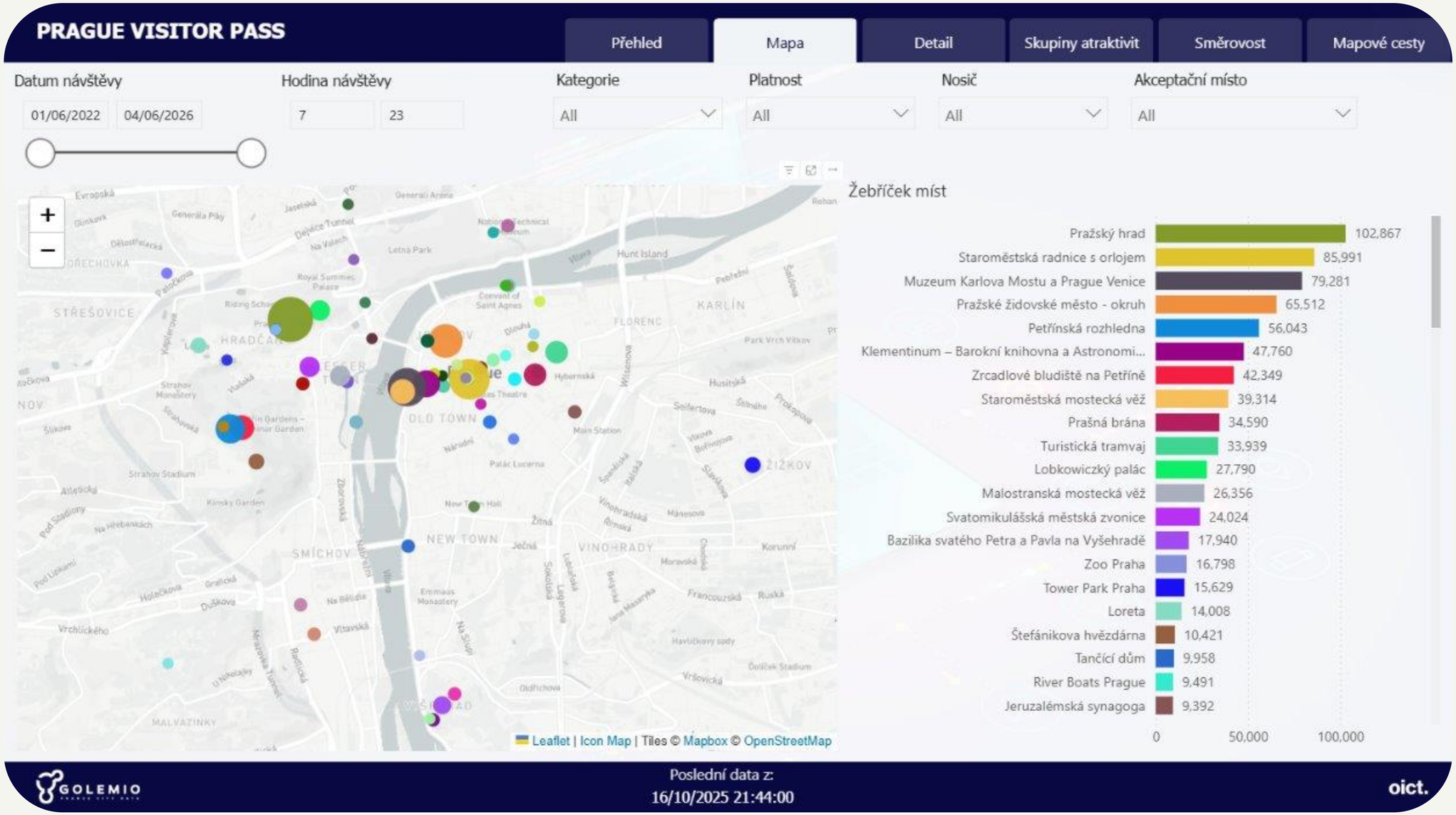
planning > review > planning

03 Architecture



04.1 Product analytics — Prague Visitor Pass

04.1 Product analytics — Prague Visitor Pass dashboard



04.2 Data-Driven Projects — Smart Waste Collection

Overview

- 7,700 Sensors
  - 3,500 for colored glass
  - 2,150 for metals
  - 1,400 for clear glass
  - 450 for paper
  - 200 for multi-commodity (plastic, Tetra Pak, ...)
- Fixed collection schedule, ranging from 7 times per week to once every 6 weeks

04.2 Data-Driven Projects — Smart Waste Collection

Fullness measurements

Better Waste Collection Suggestions

Adresa	Městská část	Druh odpadu	Svozové dny	Stávající frekvence	Objem nádoby	Změna frekvence	Navrhovaná frekvence	Počet svozů dle sensorů	Počet svozů dle Helios	Neproběhlé svozy
Africká 619/22	Praha 6	Kovy	Pá	41	1100	zvyšujeme	31	4	4	0 z 4
Akátová 1175/3	Praha 8	Kovy	Út	41	1100	zvyšujeme	31	3	3	0 z 3
Arabská 674/23	Praha 6	Barevné sklo	Pá	61	2500	zvyšujeme	51	2	2	0 z 2
Bellušova 1847/10	Praha 13	Kovy	Út	41	1100	zvyšujeme	31	3	3	0 z 3
Bohnická 320/40	Praha 8	Barevné sklo	Po	61	1100	zvyšujeme	51	2	2	0 z 2
Božanovská 2367/94	Praha 20	Kovy	Čt	41	1100	zvyšujeme	31	3	3	0 z 3
Čajkovského 1671/25	Praha 3	Kovy	St	61	3000	zvyšujeme	51	3	3	0 z 3

Reduce frequency:

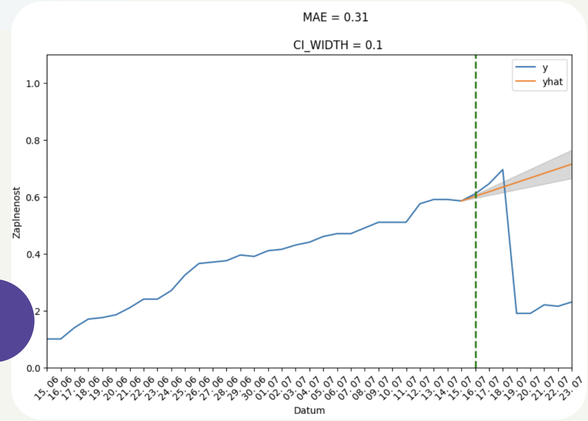
- If containers are consistently underused compared to their capacity over a period of time.
- Ensure projected fill after scheduled intervals remains below a safe threshold.

Increase frequency:

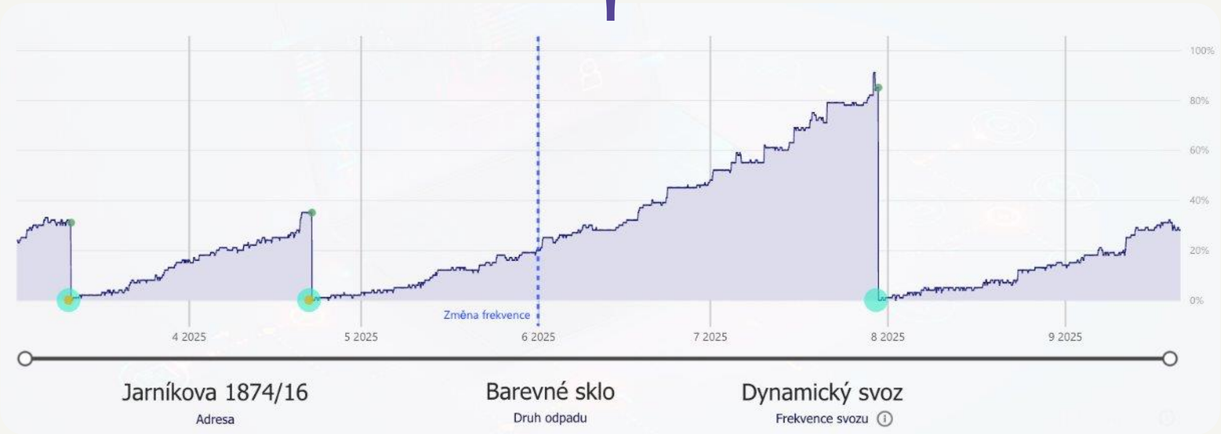
- If containers are frequently near or over capacity for multiple consecutive periods.
- Only adjust if previous scheduled collections actually occurred.

04.2 Data-Driven Projects — Dynamic Waste Colection – Introduction

Fullness measurements



prediction algorithm



When measurements Alone Is Not Enough

- A regular collection schedule is not suffiecient, even the least frequenct collection is too often
- Predicts glass container fullness and automatically issues collection orders to prevent overfilling

Goal

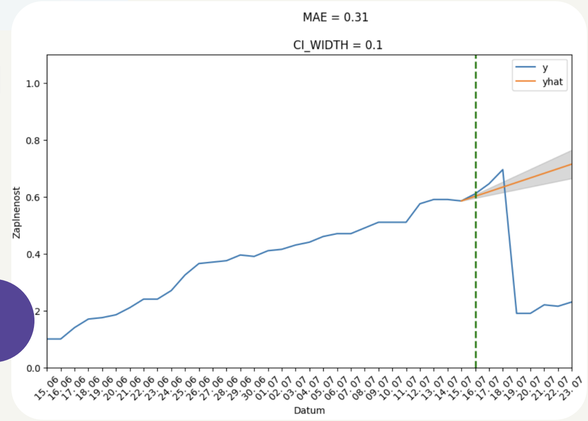
- Collect only containers that will be more than **90%** full within the next **8 days**
- PS a.s. has a week to collect them

Outputs

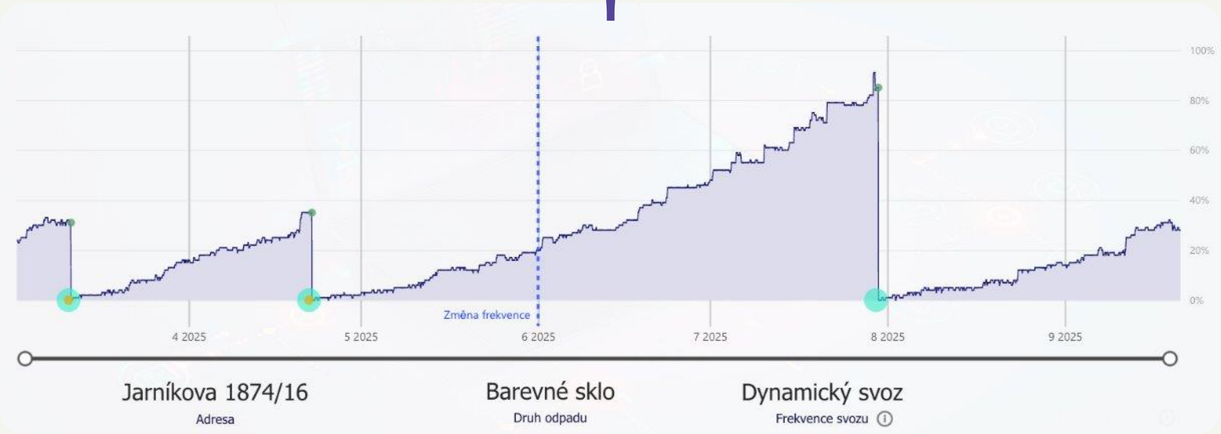
- generated instruction for PS a.s. to collect container
- Control dahboard

04.2 Data-Driven Projects — Dynamic Waste Collection – Technical Solution

Fullness measurements



prediction algorithm



Input Data

- Data from the last sensor-based collection or from the last 10% fill level
- Daily medians
- Ignores small drops which intentionally overestimate fullness
- Limited to relevant recent data

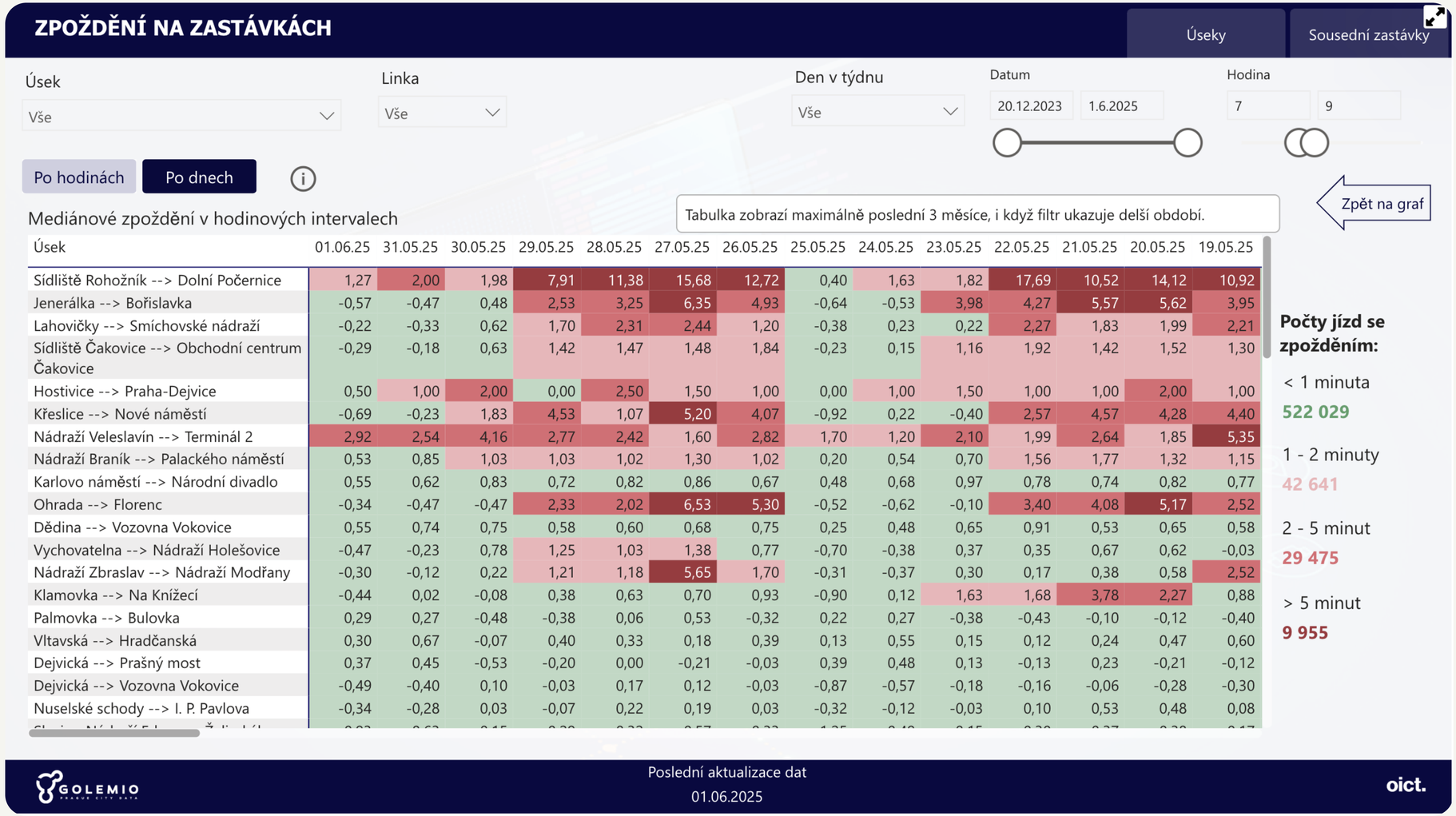
Predictive Model

- Calculation based on average daily increases/decreases
- Assumes linear growth of fill level
- Unusual cases (missing data, high uncertainty, decreasing trends) are logged.

All done in Python!

04.3 Big Data Analytics — Analyses of Delays at Public Transport Stops

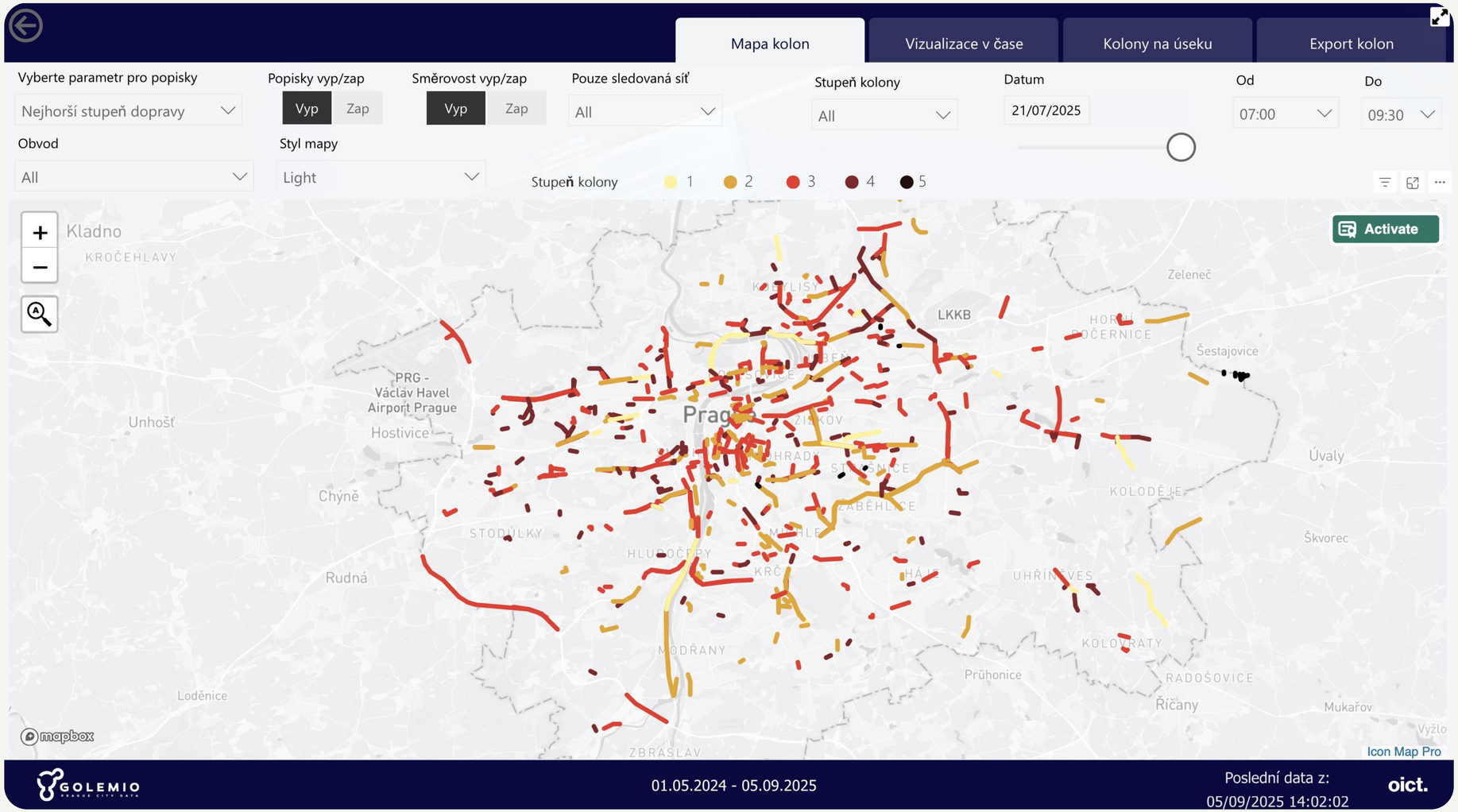
- Big data – lakehouse in Azure Blob Storage
- More data every day PID map



Úsek	01.06.25	31.05.25	30.05.25	29.05.25	28.05.25	27.05.25	26.05.25	25.05.25	24.05.25	23.05.25	22.05.25	21.05.25	20.05.25	19.05.25
Sídliště Rohožník --> Dolní Počernice	1,27	2,00	1,98	7,91	11,38	15,68	12,72	0,40	1,63	1,82	17,69	10,52	14,12	10,92
Jenerálka --> Bořislavka	-0,57	-0,47	0,48	2,53	3,25	6,35	4,93	-0,64	-0,53	3,98	4,27	5,57	5,62	3,95
Lahovičky --> Smíchovské nádraží	-0,22	-0,33	0,62	1,70	2,31	2,44	1,20	-0,38	0,23	0,22	2,27	1,83	1,99	2,21
Sídliště Čakovice --> Obchodní centrum Čakovice	-0,29	-0,18	0,63	1,42	1,47	1,48	1,84	-0,23	0,15	1,16	1,92	1,42	1,52	1,30
Hostivice --> Praha-Dejvice	0,50	1,00	2,00	0,00	2,50	1,50	1,00	0,00	1,00	1,50	1,00	1,00	2,00	1,00
Křeslice --> Nové náměstí	-0,69	-0,23	1,83	4,53	1,07	5,20	4,07	-0,92	0,22	-0,40	2,57	4,57	4,28	4,40
Nádraží Veleslavín --> Terminál 2	2,92	2,54	4,16	2,77	2,42	1,60	2,82	1,70	1,20	2,10	1,99	2,64	1,85	5,35
Nádraží Braník --> Palackého náměstí	0,53	0,85	1,03	1,03	1,02	1,30	1,02	0,20	0,54	0,70	1,56	1,77	1,32	1,15
Karlovo náměstí --> Národní divadlo	0,55	0,62	0,83	0,72	0,82	0,86	0,67	0,48	0,68	0,97	0,78	0,74	0,82	0,77
Ohrada --> Florenc	-0,34	-0,47	-0,47	2,33	2,02	6,53	5,30	-0,52	-0,62	-0,10	3,40	4,08	5,17	2,52
Dědina --> Vozovna Vokovice	0,55	0,74	0,75	0,58	0,60	0,68	0,75	0,25	0,48	0,65	0,91	0,53	0,65	0,58
Vychovatelna --> Nádraží Holešovice	-0,47	-0,23	0,78	1,25	1,03	1,38	0,77	-0,70	-0,38	0,37	0,35	0,67	0,62	-0,03
Nádraží Zbraslav --> Nádraží Modřany	-0,30	-0,12	0,22	1,21	1,18	5,65	1,70	-0,31	-0,37	0,30	0,17	0,38	0,58	2,52
Klamovka --> Na Knížecí	-0,44	0,02	-0,08	0,38	0,63	0,70	0,93	-0,90	0,12	1,63	1,68	3,78	2,27	0,88
Palmovka --> Bulovka	0,29	0,27	-0,48	-0,38	0,06	0,53	-0,32	0,22	0,27	-0,38	-0,43	-0,10	-0,12	-0,40
Vltavská --> Hradčanská	0,30	0,67	-0,07	0,40	0,33	0,18	0,39	0,13	0,55	0,15	0,12	0,24	0,47	0,60
Dejvická --> Prašný most	0,37	0,45	-0,53	-0,20	0,00	-0,21	-0,03	0,39	0,48	0,13	-0,13	0,23	-0,21	-0,12
Dejvická --> Vozovna Vokovice	-0,49	-0,40	0,10	-0,03	0,17	0,12	-0,03	-0,87	-0,57	-0,18	-0,16	-0,06	-0,28	-0,30
Nuselské schody --> I. P. Pavlova	-0,34	-0,28	0,03	-0,07	0,22	0,19	0,03	-0,32	-0,12	-0,03	0,10	0,53	0,48	0,08

04.4 Long-Term Cooperation — Waze and Other Traffic Data

- Traffic jam map



Thank you!

Contacts

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